Midterm 2 | PHYS 5460 | Solutions

Problem 1:

a) $\frac{N_0}{N} = 1 - \left(\frac{T}{T_c}\right)^{3/2} = 1 - \left(\frac{1}{2}\right)^{3/2} = 1 - \frac{1}{\sqrt{8}} = 0.65$

b) C_V is continuous but it's derivative is not so this is 2nd order.

Problem 2:

a) $N = 2\sum_k n_k = 2\frac{V}{(2\pi)^3}\int d^3k \frac{1}{e^{\frac{\hbar ck - \mu}{k_B T}} + 1}$

But at $T=0$ $\quad N = \frac{2V}{(2\pi)^3} 4\pi \int_0^{k_F} dk\, k^2$

$N = \frac{V}{\pi^2}\frac{k_F^3}{3}$ or $k_F = \left(3\pi^2 \frac{N}{V}\right)^{1/3}$

Then $\varepsilon_F = \hbar c k_F = \hbar c \left(3\pi^2 \frac{N}{V}\right)^{1/3}$

b) $E = 2\sum_k \epsilon_k n_k = \frac{2V}{(2\pi)^3}\int_0^{k_F} d^3k\, \hbar c k$

$= \frac{V}{\pi^2}\hbar c \int_0^{k_F} dk\, k^3 = \frac{\hbar c V}{\pi^2}\frac{k_F^4}{4} = \frac{\hbar c}{4\pi^2} V \left(3\pi^2 \frac{N}{V}\right)^{4/3}$

$= \frac{\hbar c}{4\pi^2}(3\pi^2)^{4/3}\frac{N^{4/3}}{V^{1/3}}$

c) $P = -\frac{\partial E}{\partial V} = \frac{1}{3}\frac{\hbar c}{4\pi^2}(3\pi^2)^{4/3}\left(\frac{N}{V}\right)^{4/3}$

$= \frac{\hbar c}{12\pi^2}(3\pi^2 n)^{4/3}$

Here we have $n^{4/3}$ dependence versus $n^{5/3}$ dependence for the non-relativistic case.